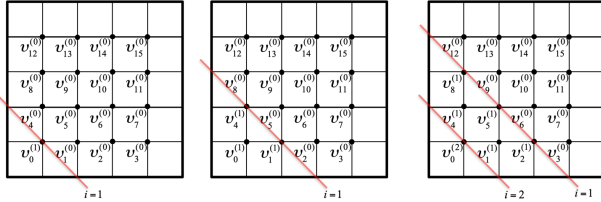




7.4.2 Parallelism in splitting methods ¶

One of the advantages of, for example, the Jacobi iteration over the Gauss-Seidel iteration is that the values at all mesh points can be updated simultaneously. This comes at the expense of slower convergence to the solution.

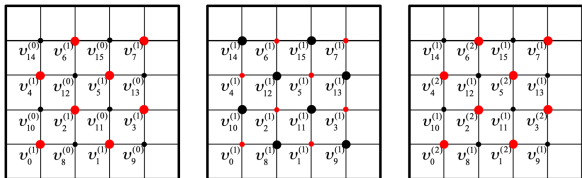
There is actually quite a bit of parallelism to be exploited in the Gauss-Seidel iteration as well. Consider our example of a mesh on a square domain as illustrated by



- First $v_0^{(1)}$ is computed from $v_1^{(0)}$ and $v_N^{(0)}$.
- Second, simultaneously,
 - $v_1^{(1)}$ can be computed from $v_0^{(1)}$, $v_2^{(0)}$, and $v_{N+1}^{(0)}$.
 - $v_N^{(1)}$ can be computed from $v_0^{(1)}$, $v_{N+1}^{(0)}$, and $v_{2N}^{(0)}$.
- Third, simultaneously,
 - $v_2^{(1)}$ can be computed from $v_1^{(1)}$, $v_2^{(0)}$, and $v_{N+2}^{(0)}$.
 - $v_{N+1}^{(1)}$ can be computed from $v_1^{(1)}$, $v_N^{(0)}$, and $v_{N+2}^{(0)}$ and $v_{2N+1}^{(0)}$.
 - $v_{2N}^{(1)}$ can be computed from $v_N^{(1)}$ and $v_{2N+1}^{(0)}$ and $v_{3N}^{(0)}$.
 - AND $v_0^{(2)}$ can be computed from $v_1^{(1)}$ and $v_N^{(1)}$, which starts a new "wave."

What we notice is that taking the opportunity to update when data is ready creates wavefronts through the mesh, where each wavefront corresponds to computation related to a different iteration.

Alternatively, extra parallelism can be achieved by ordering the mesh points using what is called a **red-black ordering**. Again focusing on our example of a mesh placed on a domain, the idea is to partition the mesh points into two groups, where each group consists of points that are not adjacent in the mesh: the red points and the black points.



The iteration then proceeds by alternating between (simultaneously) updating all values at the red points and (simultaneously) updating all values at the black points, always using the most updated values.